

Week 6 ε_t is WN $\Rightarrow E(\varepsilon_t) = 0$

Week 7 ε_t is MDS $\Rightarrow E(\varepsilon_t | F_{t-1}) = 0$ $E(\varepsilon_{t+2} | F_{t-1}) = \text{not MDS}$

gap 1-period

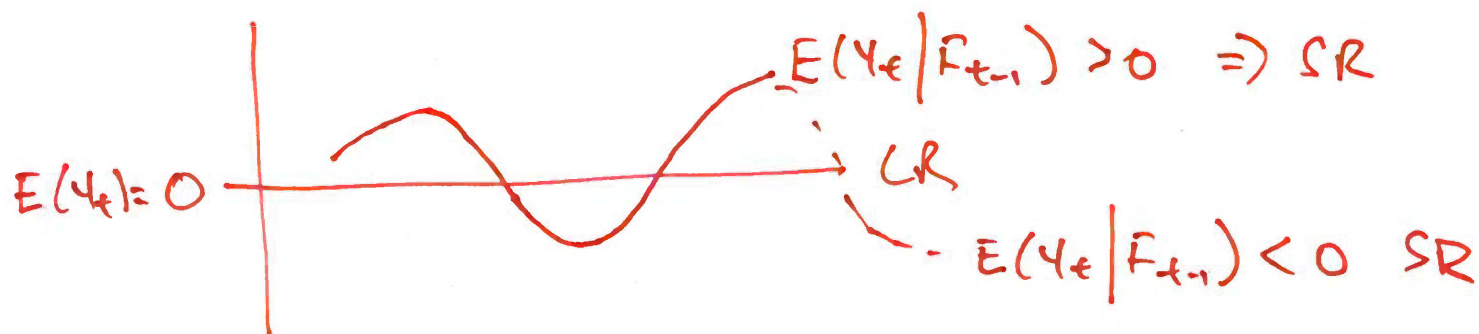
$$E(\varepsilon_t) = E(E(\varepsilon_t | F_{t-1})) \stackrel{LIE}{=} E(0) = 0$$

AR(1) $y_t = c + \phi_1 y_{t-1} + \varepsilon_t$

$$E(y_t | F_{t-1}) = c + \phi_1 y_{t-1} = \text{Short Run} = RV$$

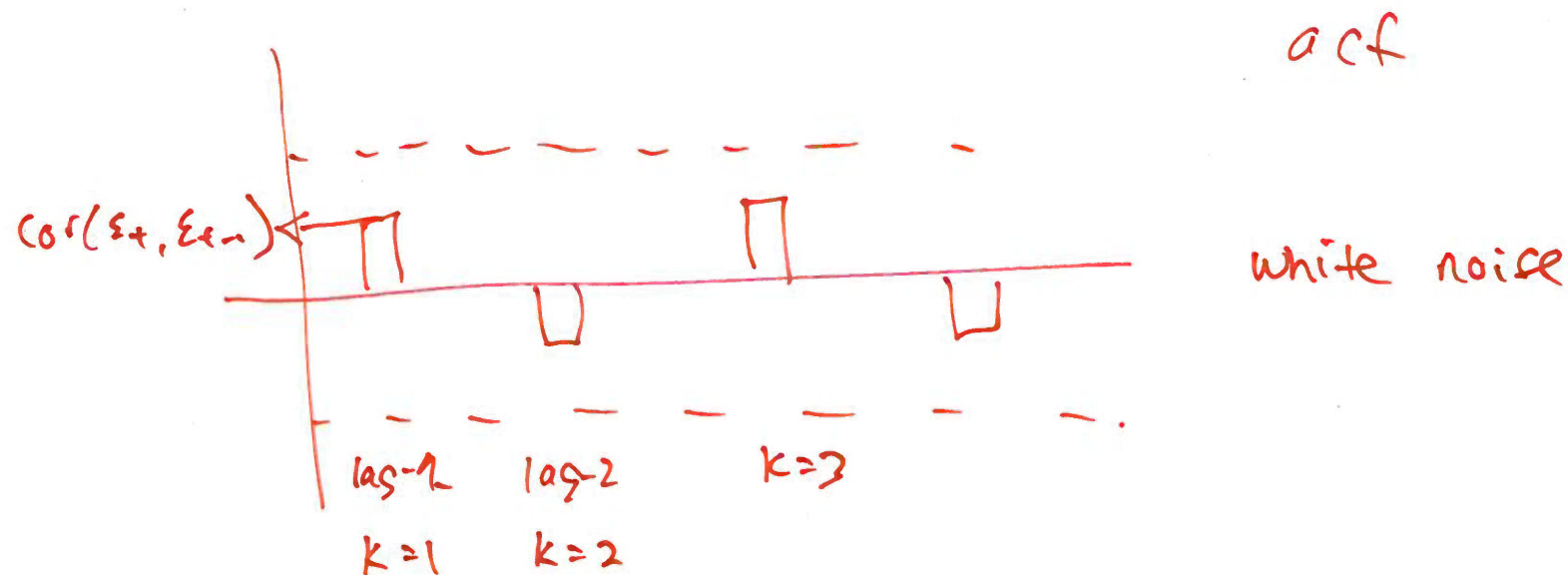
$$E(y_t) = \frac{c}{1 - \phi_1} = \text{Long Run} = \text{a number}$$

$c = 0$



ε_t is WN $\Rightarrow$ no autocorrelation
 $\Rightarrow$ no serial correlation
no remembers
forget

$\left. \begin{array}{l} \Rightarrow \text{no autocorrelation} \\ \Rightarrow \text{no serial correlation} \end{array} \right\} \text{Corr}(\varepsilon_t, \varepsilon_{t-k}) = 0$
 $k=1, 2, 3, 4, \dots$



$$\hookrightarrow E(a Y_t) = a E(Y_t)$$

$$Y_t \equiv R \cdot V.$$

a known constant = 0.5

$$\hookrightarrow E(a Y_t) \neq Y_t E(a)$$

$$\hookrightarrow E(\sigma_t u_t | F_{t-1}) = \sigma_t E(u_t | F_{t-1})$$

σ_t is known given F_{t-1}

$$\varepsilon_t = \sigma_t u_t$$

LIE

$$E(\varepsilon_t) = E(E(\varepsilon_t | F_{t-1}))$$

$$= E(\sigma_t u_t | F_{t-1})$$

$$= \sigma_t E(u_t | F_{t-1})$$

$$= 0$$

σ_t is known
given F_{t-1}

Slide 42/43.

$$\text{cov}(\xi_t, \xi_{t-k}) = E(\xi_t - E(\xi_t))(\xi_{t-k} - E(\xi_{t-k})) \quad k=1,2,3.$$

$$= E(\xi_t \xi_{t-k})$$

$$= E\left(E(\xi_t \xi_{t-k} | F_{t-1})\right) \quad \text{LIE}$$

$$= E\left(\xi_{t-k} \underbrace{E(\xi_t | F_{t-1})}_{\text{slide 42}}\right)$$

ξ_{t-1} is
known

$$\xi_{t-1} = 10$$

$$= E(\xi_{t-k} \times 0)$$

$$= E(0) = 0$$

$$V(\xi_t) = E(\xi_t - E(\xi_t))^2$$

$$= E(\xi_t^2)$$

$$= E\left(E(\xi_t^2 | F_{t-1})\right)$$

$$= E(\xi_t^2) \quad \text{slide 43}$$

$$= E\left(\underbrace{\alpha_0}_{\text{ARCH(1)}} + \underbrace{\alpha_1 \xi_{t-1}^2}_{\text{ARCH(1)}}\right)$$

$$= E(\alpha_0) + E(\alpha_1 \xi_{t-1}^2)$$

$$= \alpha_0 + \alpha_1 E(\xi_{t-1}^2)$$

$$= \alpha_0 + \alpha_1 E(\xi_t^2)$$

$$= \alpha_0 + \alpha_1 V(\xi_t)$$

$$V(\xi_t) = \frac{\alpha_0}{1 - \alpha_1}$$